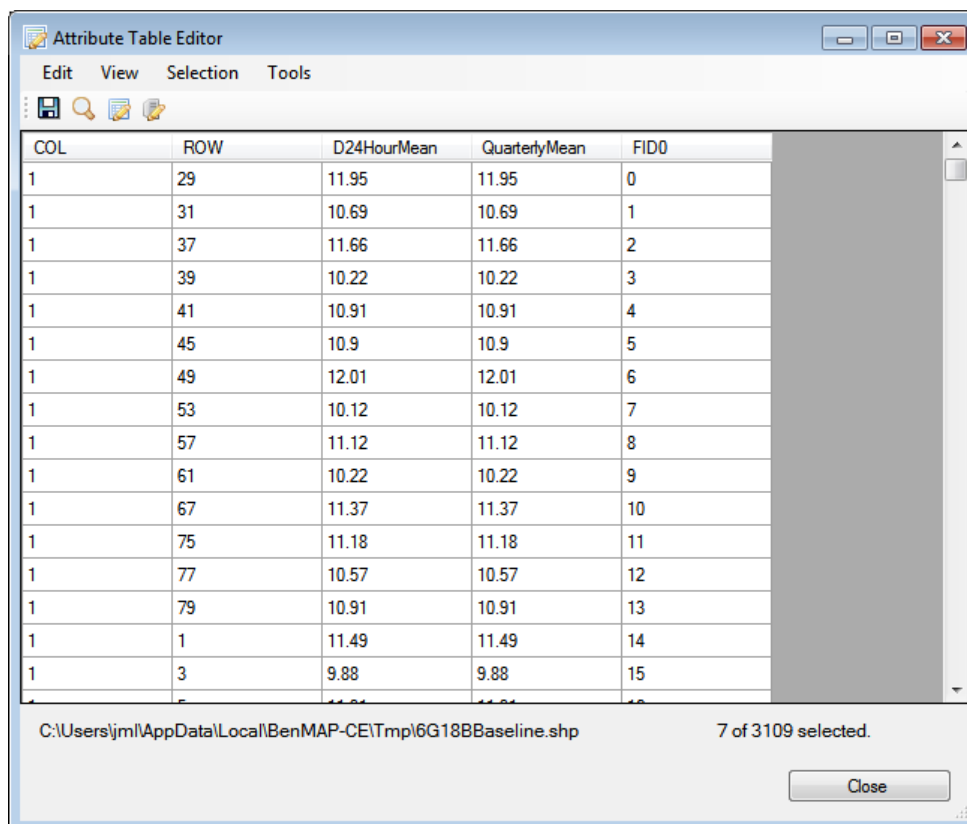
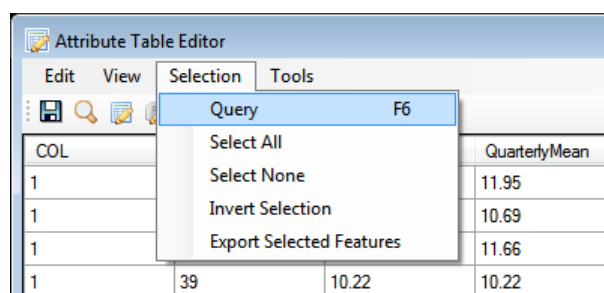


Creating a Query

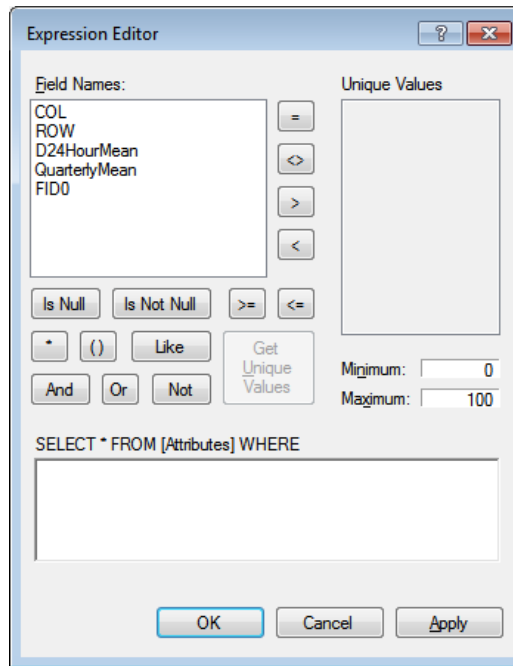
You can create a query on any map layer that shows up in the GIS table of contents. After importing a file (such as **Baseline** or **Control**) you can begin a query by selecting a layer. The selected layer will become highlighted with a light blue bubble around it. Next, click the **Attribute Table** button in the GIS toolbar. This will open the **Attribute Table Editor** window, shown below:



From the window above, click **Selection** from the top menu bar. This will open a drop-down menu where you can select the **Query** option.

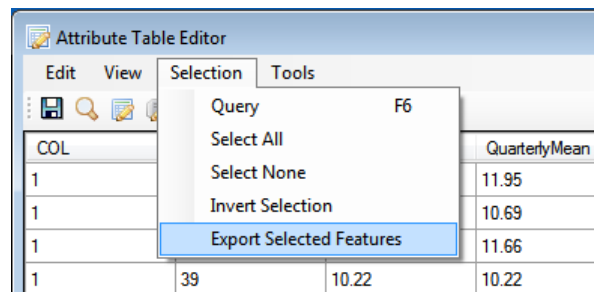


This will open **Expression Editor** window (shown below):



The **Expression Editor** is used to query your data. First, select a **Field Name** to query, by double-clicking on a desired entry. The **Field Name** should show up in brackets in the bottom text box. Next, select an operator from the panel of buttons. The selected operator symbol should show up in the bottom text box next to the field name. Finally, to complete the first query entry, click in the bottom text box (after the operator symbol) and enter a value that you would like to compare against (e.g., `[D24HourMean] >= 15`). More attributes can be added to the query by clicking the **And**, **Or**, or **Not** buttons. Once you are satisfied with the query statement, click **OK** on the **Expression Editor** window. It may take a few minutes to find all the results. The **Attributes Table Editor** should appear, with the requested selections highlighted. The map on the GIS portion of the main BenMAP-CE home screen should show the query results.

To save these results, click the **Selection** button from the top menu bar of the **Attribute Table Editor** window. From the dropdown menu, select the **Export Selected Features** entry. This will open a **Save As** window, where you can save the query into a shapefile (.shp). Click the **Close** button on the **Attribute Table Editor**.



To add the query as a new layer, click the ‘+’ button located in the GIS toolbar (located next to the GIS table of contents). An **Open** window will be displayed, where you can select your recently saved shapefile. Select the file and click **Open**. The new layer should appear near the top of the GIS table of contents. The layer can be dragged-and-dropped into any map group you would like within the GIS table of contents.

8.3.2 Data Tab

The **Data** tab allows you to view all the data that is being presented in the map.

The screenshot shows the BenMAP-CE 1.4 Data tab. The left sidebar contains a tree view with the following structure:

- Air Quality Surfaces
 - Pollutant
 - Source of Air Quality Data (PM2.5)
 - Baseline
 - Base Data
 - Control
 - Control Data
 - Air quality delta (baseline - control)
- Estimate Health Impacts
 - Population Dataset
 - Health Impact Functions
- Aggregate, Pool & Value
 - Aggregation
 - Pooling Method
 - Valuation Method

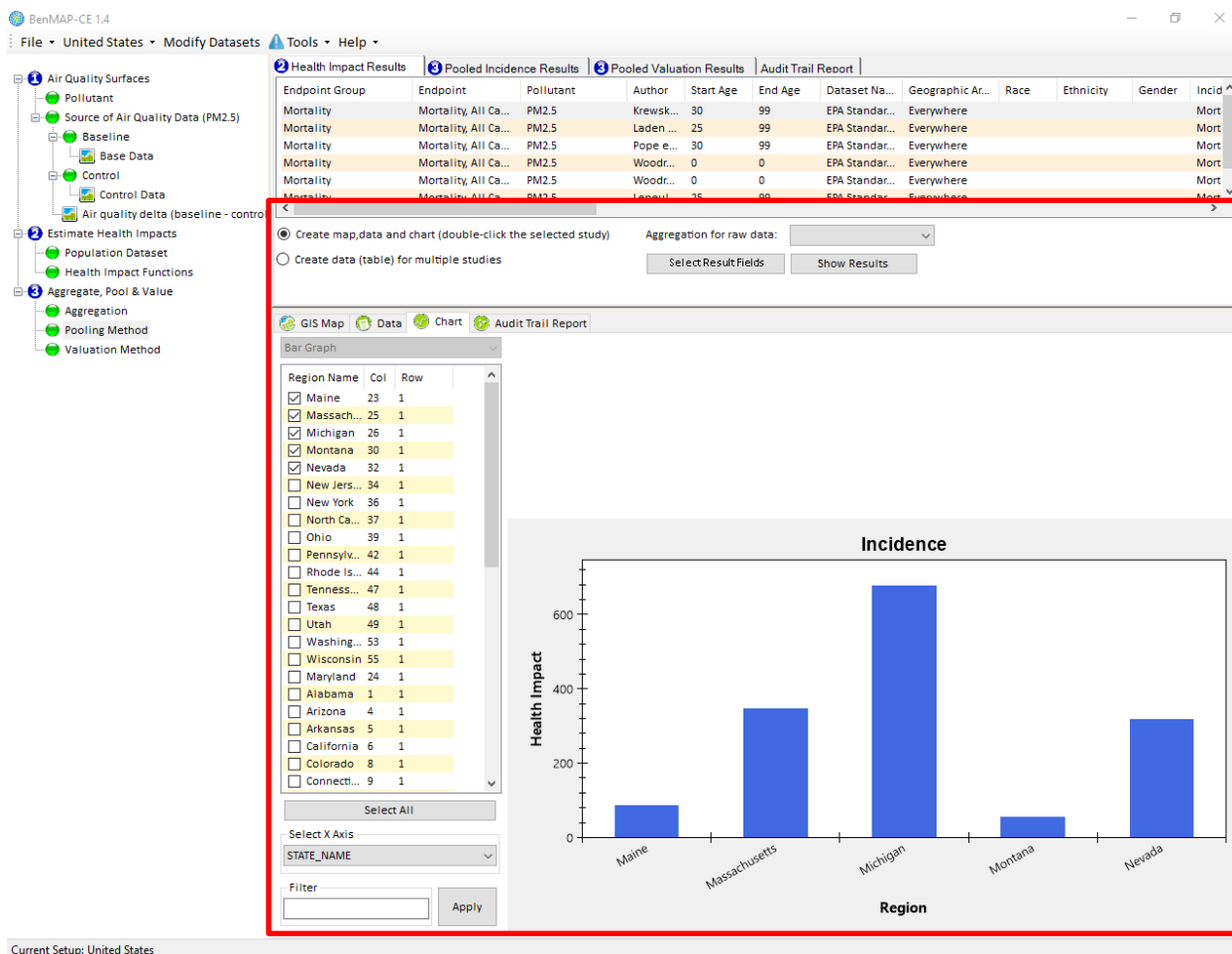
The main data table has the following columns: Column, Row, Endpoint, Author, Start Age, End Age, Point Estimate, Population, Delta, Mean, Baseline, Percent Of Baseline, and Start. The table displays 34 rows of data. At the bottom of the table, there are navigation buttons (back, forward, etc.), a 'Digits After Decimal Point' dropdown set to 4, and an 'Export...' button.

At the bottom of this tab, there are a few options:

- The left-most set of buttons allows you to toggle between pages of data.
- The middle option allows you to change the number of digits that appear after the decimal point.
- The far right entry allows you to export the data table. Clicking the **Export** button opens a **Save As** window, allowing you to save the data as a .csv file.

8.3.3 Chart Tab

The **Chart** tab allows you to select certain regions of data within the main layer to compare localized results more easily.



You have the option of selecting the regions that you would like to compare using the list to the left of the chart by checking and unchecking certain regions. The chart automatically updates with each new selection.²⁶

8.3.4 Audit Trail Report

Audit Trail Reports facilitate transparency and reproducibility by reporting a summary of your assumptions underlying each of five types of files generated by BenMAP-CE: Air Quality Grids (with the *.aggx* extension), Incidence Configurations (with the *.cfgx* extension), Configuration Results (with the *.cfgrx* extension), Aggregation, Pooling, and Valuation Configurations (with the *.apvx* extension), and Aggregation, Pooling, and Valuation Results (with the *.apvr* extension).

²⁶ The data charts were originally developed using ZedGraph software, which is no longer supported. The BenMAP-CE development team is considering options to update and improve this feature.